RH

QQRH ALMIGHTY PUSH AWPI CAYLEY GOLPHI
TAYLORS PRODIGY STIRLINGS PRODIGY MISTERPREMISE
GIGGA SMALL O SOLOBOLO BKQR DIRICHLET

0 RIEMANN

TARGET / DESCENT / GEOMETRY

$$\xi(s) := \frac{1 + s(s-1)\zeta_0^*(s)}{2}, \qquad \Xi(w) := \xi\left(\tfrac{1}{2} + \mathrm{i}w\right), \qquad \Theta(z) := \xi\left(\tfrac{1}{2} + \sqrt{z}\right). \tag{0.1}$$

$$c_r(x) := r\left(x + \mathrm{i}\sqrt{1 - x^2}\right), \quad P(r,x) := |\Theta(c_r(x))|^2, \quad \mathscr{U} := (0,\infty) \times (-1,1). \tag{0.2}$$

$$A(r,x) := \partial_x P(r,x), \qquad \mathsf{A} :\Longleftrightarrow \forall (r,x) \in \mathscr{U}, \quad A(r,x) \geq 0. \tag{0.3}$$

$$\boxed{\mathrm{RH} \iff \mathsf{A} \iff [\Theta^{-1}(0) \subset (-\infty, 0]] \iff [\Xi^{-1}(0) \subset \mathbb{R}].} \tag{0.4}$$

CRITICAL STRIP / SQUARED COORDINATES

$$s = \tfrac{1}{2} + d + \mathrm{i}t, \quad d^2 + t^2 > 0, \quad z = (s - \tfrac{1}{2})^2, \quad r = d^2 + t^2, \quad x = \frac{d^2 - t^2}{d^2 + t^2}. \tag{0.5}$$

$$d^2 = \frac{r(1+x)}{2}, \qquad t^2 = \frac{r(1-x)}{2}, \qquad |d| < \tfrac{1}{2} \iff r(1+x) < \tfrac{1}{2}. \tag{0.6}$$

$$\begin{gathered} d = 0,\ t \neq 0 \Longrightarrow x = -1,\ r = t^2; \\ P(r,x) \geq 0, \qquad P(r,x) > 0 \iff \Theta(c_r(x)) \neq 0. \end{gathered} \tag{0.7}$$

1 QQRH

OPERATION: ALMIGHTY PUSH / COVERAGE / EXHAUSTION

$$Q(m) :\Longleftrightarrow \forall s, \quad \left[0 < \operatorname{Re} s < 1,\ |\operatorname{Im} s| < m|\operatorname{Re} s - \tfrac{1}{2}|\right] \Longrightarrow \zeta(s) \neq 0. \tag{1.1}$$

$$a_m := \frac{1-m^2}{1+m^2}, \qquad a_2 = -\frac{3}{5}, \quad a_{11} = -\frac{60}{61}, \quad a_{13} = -\frac{84}{85}. \tag{1.2}$$

$$m > 0, \quad |t| < m|d| \Longrightarrow \frac{\operatorname{Re}((d+\mathrm{i}t)^2)}{|(d+\mathrm{i}t)^2|} = \frac{d^2-t^2}{d^2+t^2} > a_m. \tag{1.3}$$

$$m > 0, \quad [\forall r > 0,\ a_m < x < 1 :\ A(r,x) \geq 0] \Longrightarrow Q(m). \tag{1.4}$$

ANCHOR / THIRTEEN

$$R_* := \frac{255}{7}, \qquad a_* := -\frac{84}{85}, \qquad Q(2),\ Q(10),\ Q(11),\ Q(13). \tag{1.5}$$

$$\forall (r,x) \in \mathscr{U} :\quad [r \leq R_* \ \vee\ x > a_*] \Longrightarrow A(r,x) \geq 0. \tag{1.6}$$

$$0 \leq r \leq R_*, \quad -1 \leq x \leq 1 \Longrightarrow P(r,x) > 0. \tag{1.7}$$

RESIDUAL FRONTIER / OPEN

$$\mathsf{R} :\Longleftrightarrow \forall r > R_*,\ \forall -1 < x \leq a_* :\ A(r,x) \geq 0, \qquad \boxed{\mathsf{R} \iff \mathsf{A} \iff \mathrm{RH}.} \tag{1.8}$$

COFINAL SLOPES

$$\boxed{\mathrm{RH} \iff [\forall m > 0,\ Q(m)] \iff [\forall M > 0\ \exists m \geq M :\ Q(m)].} \tag{1.9}$$

$$\mathrm{RH} \iff [\forall m > 0,\ \forall r > 0,\ \forall a_m < x < 1 :\ A(r,x) \geq 0]\,. \tag{1.10}$$

2 ANGULAR REGULARITY

STATIONARY / MASTER / LAGARIAS

$$\Theta \text{ entire} \Longrightarrow P(r,\cdot) \in C^\infty((-1,1)), \qquad P(r,x) = 0 \Longrightarrow A(r,x) = 0. \tag{2.1}$$

$$\mathsf{N} :\Longleftrightarrow \forall (r,x) \in \mathscr{U} : \ P(r,x) > 0 \Longrightarrow A(r,x) \neq 0. \tag{2.2}$$

$$\mathsf{S} :\Longleftrightarrow \forall (r,x) \in \mathscr{U} : \ A(r,x) = 0 \Longrightarrow P(r,x) = 0. \tag{2.3}$$

$$\boxed{\mathsf{N} \iff \mathsf{S}, \qquad \mathsf{N} \Longrightarrow \mathrm{RH}, \qquad \mathsf{S} \Longrightarrow \mathrm{RH}.} \tag{2.4}$$

$$\neg\mathrm{RH} \Longrightarrow \exists (r,x) \in \mathscr{U} : \ P(r,x) > 0, \quad A(r,x) = 0. \tag{2.5}$$

PROFILE MASTER

$$I := [-1,1], \quad D_I f(x) := \mathrm{derivWithin}(f, I, x), \quad U(r,x) := D_I(P(r,\cdot))(x). \tag{2.6}$$

$$\mathsf{PM} :\Longleftrightarrow [\forall r > 0, \ x \in [-1,1] : \ U(r,-1) \le U(r,x)] \wedge [\forall r > 0 : \ U(r,-1) \ge 0]\,. \tag{2.7}$$

$$L_\varepsilon(r,x) := \log\!\left(1 + \frac{\varepsilon P(r,x)}{|\Theta(r)|^2}\right), \quad J_\varepsilon := e^{L_\varepsilon} D_I L_\varepsilon, \quad \mathsf{FM}_\varepsilon := \mathsf{PM}[U \mapsto J_\varepsilon]. \tag{2.8}$$

$$\boxed{\varepsilon > 0 \Longrightarrow (\mathrm{RH} \iff \mathsf{PM} \iff \mathsf{FM}_\varepsilon).} \tag{2.9}$$

$$a/0 := 0, \qquad \Theta(r) = 0 \Longrightarrow L_\varepsilon(r,\cdot) = 0 \Longrightarrow J_\varepsilon(r,\cdot) = 0. \tag{2.10}$$

LAGARIAS / STRICT SIGN

$$\boxed{\begin{aligned} \mathrm{RH} &\iff \left[\forall\, \mathrm{Re}\, s > \tfrac{1}{2} : \ \xi(s) \neq 0, \ \mathrm{Re}\,\frac{\xi'(s)}{\xi(s)} > 0\right] \\ &\iff \left[\forall\, \mathrm{Im}\, w > 0 : \ \mathrm{Im}\,\frac{\Xi'(w)}{\Xi(w)} < 0\right]. \end{aligned}} \tag{2.11}$$

3 TAYLOR / STIRLING

TAYLORS PRODIGY / STIRLINGS PRODIGY

$$H(r,x) := A(r,x), \qquad F(r,\theta) := H(r,\cos\theta), \qquad 0 < \theta < \pi. \tag{3.1}$$

TAYLORS PRODIGY

$$\begin{aligned}\mathcal{K}_r := \{k : \mathbb{R} \to [0,\infty) : \quad & \forall b \in (0,\pi),\ k \in L^1([0,b]), \\ & \forall \theta \in (0,\pi),\ F(r,\theta) > 0 \\ & \qquad \Longrightarrow F_\theta(r,\theta) \geq -k(\theta)F(r,\theta)\}.\end{aligned} \tag{3.2}$$

$$\mathsf{TP} :\Longleftrightarrow \forall r > 0 : \mathcal{K}_r \neq \varnothing, \qquad \boxed{\mathsf{TP} \iff \mathrm{RH}.} \tag{3.3}$$

$$\begin{aligned}& r > 0,\ k \in \mathcal{K}_r,\ 0 < a \leq b < \pi,\ F(r,a) > 0 \\ & \Longrightarrow F(r,b) \geq F(r,a)\exp\left(-\int_a^b k(t)\,\mathrm{d}t\right) > 0.\end{aligned} \tag{3.4}$$

$$\mathsf{TP} \Longrightarrow [H(r,y) > 0 \Longrightarrow H(r,x) > 0]_{-1<x\leq y<1}. \tag{3.5}$$

STIRLINGS PRODIGY / LOCAL

$$\begin{aligned}\mathsf{SL} :\Longleftrightarrow \forall (r,x) \in \mathscr{U},\ & \forall \epsilon > 0\ \exists \eta > 0\ \forall h \in \mathbb{R} : \\ & 0 < h < \eta,\ x + h \leq 1 \Longrightarrow P(r,x+h) - P(r,x) \geq -\epsilon h.\end{aligned} \tag{3.6}$$

$$\boxed{\mathsf{SL} \iff \mathsf{A} \iff \mathrm{RH}.} \tag{3.7}$$

STIRLINGS PRODIGY / GLOBAL / CONDITIONAL

$$U := D_I P, \qquad a(r) := U(r,1), \qquad b(r) := D_I(U(r,\cdot))(1). \tag{3.8}$$

$$\begin{aligned}\mathsf{SG} :\Longleftrightarrow \forall r > 0 : \quad & [\forall -1 < x < 1 :\ U(r,x) \geq a(r) - b(r)(1-x)]\,, \\ & 2b(r) \leq a(r), \qquad b(r) > 0.\end{aligned} \tag{3.9}$$

$$\mathsf{SG} \Longrightarrow U(r,x) \geq b(r)(1+x) > 0 \Longrightarrow \mathsf{SL} \Longrightarrow \mathrm{RH}. \tag{3.10}$$

4 MISTERPREMISE

GIGGA SMALL O / WEIGHTED TRANSPORT / TRACE

$$\kappa(r) := \inf_{R \ge 85/2} \frac{(R-\frac{1}{2})(R^2+r^2)}{2rR^2}, \qquad r > R_* = \frac{255}{7}. \tag{4.1}$$

$$\begin{aligned} \mathsf{MP} :\Longleftrightarrow\ & \forall r > R_*\ \exists M > 0\ \forall -1 < x \le -\kappa(r) : \\ & A(r,x) > 0 \Longrightarrow (1+x)A_x(r,x) \le M A(r,x). \end{aligned} \tag{4.2}$$

$$\boxed{\mathsf{MP} \iff \mathrm{RH.}} \qquad M = M(r) \in (0,\infty). \tag{4.3}$$

$$\mathsf{MP},\ A(r,y) > 0 \Longrightarrow A(r,t) > 0 \quad (-1 < t \le y \le -\kappa(r)). \tag{4.4}$$

GIGGA SMALL O

$$I(r) := \int_{-1}^{1} \frac{A(r,x)}{P(r,x)} \sqrt{1-x^2}\,\mathrm{d}x, \qquad q_1 := \operatorname{Re} \frac{\Theta'(0)}{\Theta(0)}. \tag{4.5}$$

$$S_{\mathrm{line}}(T) := \sum_{\substack{0<\gamma\le T \\ \zeta(1/2+\mathrm{i}\gamma)=0}} \frac{m_\gamma}{\gamma^2}, \qquad \lim_{r\downarrow 0} \frac{I(r)}{r} = \pi q_1. \tag{4.6}$$

$$\mathsf{GSO} :\Longleftrightarrow I(r) - \pi r S_{\mathrm{line}}(1/r) = o(r) \quad (r \downarrow 0). \tag{4.7}$$

$$\boxed{\mathsf{GSO} \iff \mathrm{RH.}} \qquad \mathsf{GSO} \Longrightarrow S_{\mathrm{line}}(1/r) \longrightarrow q_1. \tag{4.8}$$

DIVISOR / TRACE

$$\operatorname{div}_0 \Xi = \sum_{i \in \mathcal{I}} ([w_i] + [-w_i]), \qquad w_i \ne 0. \tag{4.9}$$

$$\omega_i := \operatorname{Re}(w_i^{-2}), \qquad q_1 = \sum_i \omega_i, \qquad \omega_i > 0, \qquad \sum_i \omega_i < \infty. \tag{4.10}$$

$$\Im w_{i_0} \ne 0 \Longrightarrow 0 < \omega_{i_0} \le q_1 - S_{\mathrm{line}}(1/r) \quad (r > 0). \tag{4.11}$$

$$\mathsf{GSO} \Longrightarrow q_1 - S_{\mathrm{line}}(1/r) \to 0 \Longrightarrow \forall i \in \mathcal{I} : \Im w_i = 0. \tag{4.12}$$

5 AWPI / PHIRH

PHI / KQ / ONE-SIDED PASSIVITY

$$\rho=\sigma+\mathrm{i}\tau,\quad \mathcal{Z}(\rho):\Longleftrightarrow[\zeta(\rho)=0,\ \rho\notin-2\mathbb{N}_{>0},\ \rho\neq1],\quad g(\rho):=2^{2(\sigma-1/2)}. \tag{5.1}$$

$$\Phi(s):=\sum_{k\geq0}4\,2^{-s}\big((2k+1)^{-s}-(2k+2)^{-s}\big). \tag{5.2}$$

$$\operatorname{Re}s>1\Longrightarrow\Phi(s)=4\,2^{-s}(1-2^{1-s})\zeta(s),\qquad \mathcal{Z}(\rho)\Longrightarrow\Phi(\rho+1)\neq0. \tag{5.3}$$

$$J(\rho):=|\rho|^2|\Phi(\rho+1)|^2>0\quad(\mathcal{Z}(\rho)),\qquad \Delta_+(\rho):=(g(\rho)-1)J(\rho)^2. \tag{5.4}$$

PLUS PASSIVITY / INPUT

$$\mathsf{W}:\Longleftrightarrow\forall\rho,\ \mathcal{Z}(\rho)\Longrightarrow\Delta_+(\rho)\leq0,\qquad \boxed{\mathsf{W}\iff\mathrm{RH}.} \tag{5.5}$$

KQ / EXACT-TO-TAIL / INPUT

$$\begin{aligned}\phi_n(s)&:=4(-1)^n(2n+2)^{-s},\\ q_n(\sigma,\tau)&:=-\Re\big(e^{-\mathrm{i}\tau\log2}\phi_n(\sigma+\mathrm{i}\tau)\big).\end{aligned} \tag{5.6}$$

$$q_n^2=16(2n+2)^{-2\sigma}\cos^2\big(\tau\log(4n+4)\big)\leq16(2n+2)^{-2\sigma}\longrightarrow0\quad(\sigma>0). \tag{5.7}$$

$$\mathsf{K}:\Longleftrightarrow\forall n,\rho,\ \mathcal{Z}(\rho)\Longrightarrow\Delta_+(\rho)\leq q_n(\sigma,\tau)^2,\qquad \boxed{\mathsf{K}\iff\mathsf{W}\iff\mathrm{RH}.} \tag{5.8}$$

$$\begin{aligned}&\Delta_+\leq q_n^2\ \forall n\Longrightarrow\Delta_+\leq0,\\ &\mathsf{W}\Longrightarrow[\sigma\leq\tfrac12,\ 1-\sigma\leq\tfrac12]_{\mathcal{Z}(\rho)}\Longrightarrow\mathrm{RH}.\end{aligned} \tag{5.9}$$

MINIMAL FRONTIER

$$J_0(\rho):=|\rho|^2,\qquad \boxed{\mathrm{RH}\iff[\forall\mathcal{Z}(\rho):g(\rho)\leq1]\iff[\forall\mathcal{Z}(\rho):\sigma\leq\tfrac12].} \tag{5.10}$$

6 CAYLEY / GOLPHI

ZERO LOSS / SATURATION / PRODUCT GEOMETRY

$$g(\sigma) := 2^{2(\sigma-1/2)}, \qquad D(\sigma) := \left(\frac{g(\sigma)-1}{g(\sigma)+1}\right)^2, \qquad T(\sigma) := 1 - D(\sigma). \tag{6.1}$$

$$E(u) := u^2, \quad C_\sigma u := \sqrt{T(\sigma)}\, u, \qquad E(u) - E(C_\sigma u) = D(\sigma) u^2. \tag{6.2}$$

$$D(\sigma) \geq 0, \qquad \boxed{D(\sigma) = 0 \iff T(\sigma) = 1 \iff \sigma = \tfrac{1}{2}.} \tag{6.3}$$

LITERAL ZERO SUPPORT

$$G(s) := \mathrm{GolPhi}^*_{S1}(s) = \xi(s), \qquad \mathcal{Z}_G(s) :\Longleftrightarrow G(s) = 0, \quad 0 < \mathrm{Re}\, s < 1. \tag{6.4}$$

$$\mathcal{Z}_G(s) \iff \mathcal{Z}(s), \qquad \mathrm{RH} \iff [\forall \mathcal{Z}_G(s) : \mathrm{Re}\, s = \tfrac{1}{2}]. \tag{6.5}$$

$$\boxed{\begin{aligned} \mathrm{RH} &\iff [\forall \mathcal{Z}(s) : D(\mathrm{Re}\, s) = 0] \\ &\iff [\forall \mathcal{Z}_G(s) : D(\mathrm{Re}\, s) = 0] \\ &\iff [\forall \mathcal{Z}_G(s) : T(\mathrm{Re}\, s) = 1] \\ &\iff [\forall \mathcal{Z}_G(s) : E(C_{\mathrm{Re}\, s} 1) = E(1)]. \end{aligned}} \tag{6.6}$$

SYMMETRIC PRODUCT / OFF-RAY NONVANISHING

$$q_\pm(w) := 1 - 2w \pm \sqrt{(1-2w)^2 - 1}, \qquad S(w) := \frac{\Theta(q_+(w))\Theta(q_-(w))}{\Theta(1)^2}. \tag{6.7}$$

$$\mathsf{OffRay}(w) :\Longleftrightarrow \neg(\mathrm{Im}\, w = 0 \ \wedge \ 1 \leq \mathrm{Re}\, w). \tag{6.8}$$

$$\boxed{[\forall w, \ \mathsf{OffRay}(w) \Longrightarrow S(w) \neq 0] \Longrightarrow \mathrm{RH}.} \tag{6.9}$$

7 SOLOBOLO / BKQR

MERGER 01 / 02 / 04 / 05

NATIVE

$$\xi_{\mathrm{SOLOBOLO}} = \xi_{\mathrm{MERGER02}} = \xi_{\mathrm{WIDDER}} = \xi, \qquad \Theta_{\mathrm{SOLOBOLO}} = \Theta_{\mathrm{MERGER02}} = \Theta. \tag{7.1}$$

MERGER 01 / AWPI ROW TRANSPORT

$$\begin{gathered} \mathsf{US}_{\mathrm{AWPI}} \wedge \mathsf{B}_{\mathrm{coeff}} \Longrightarrow \mathrm{RH}, \\ \mathsf{US}_{\mathrm{AWPI}} \wedge \mathsf{B}_{\mathrm{ODE}} \Longrightarrow \mathrm{RH}. \end{gathered} \tag{7.2}$$

MERGER 02 / FINITE FILTER

$$\|F_N - B\lambda^N\| \le M r_0^N, \quad B \neq 0, \quad 1 < |\lambda|, \quad 0 \le r_0 < |\lambda|. \tag{7.3}$$

$$\boxed{\mathsf{US}_{\mathrm{row}} \wedge \mathsf{B}_{\mathrm{filter}} \Longrightarrow \mathrm{RH}.} \tag{7.4}$$

MERGER 04 / LOCAL POLE

$$\begin{gathered} \neg\mathrm{RH} \Longrightarrow \exists p \in \mathscr{P}^{\mathrm{simple}}_{\mathrm{Cayley}}, \\ \mathsf{LP} := \forall p \in \mathscr{P}^{\mathrm{simple}}_{\mathrm{Cayley}},\ \exists K\ \forall N \ge K\ \exists a \in \mathscr{A}_{\mathrm{reg}} : \mu_N(a) < 0. \end{gathered} \tag{7.5}$$

MERGER 05 / COFINAL MOMENTS / FINITE TRACE

$$\mathsf{CP} :\Longleftrightarrow \forall K\ \exists N \ge K\ \forall a > 0 :\ \ \mu_N(a) \ge 0. \tag{7.6}$$

$$\boxed{\mathsf{LP} \wedge \mathsf{CP} \Longrightarrow \mathrm{RH}.} \qquad \mathsf{Channel} \wedge \mathsf{CP} \Longrightarrow \mathrm{RH}. \tag{7.7}$$

$$S = (d_M, \lambda_{M,a,j}), \qquad \forall M, a, j < d_M : \quad 0 \le \lambda_{M,a,j} \le 1. \tag{7.8}$$

$$\begin{aligned} \mathsf{Trace}(S) :\Longleftrightarrow & \forall K\ \exists k \ge K\ \forall a > 0 : \\ & \lim_{M\to\infty} \sum_{j<d_M} \lambda_{M,a,j}^{2^k} = \mu_{2^k-1}(a). \end{aligned} \tag{7.9}$$

$$\boxed{\mathsf{LP} \wedge \mathsf{Trace}(S) \Longrightarrow \mathrm{RH}.} \qquad \mathsf{Channel} \wedge \mathsf{Trace}(S) \Longrightarrow \mathrm{RH}. \tag{7.10}$$

SOLOBOLO / MODEL

$$\begin{aligned} \mathsf{CommonMoment}(T, Q) &\Longrightarrow \mathrm{RH}_{T.\mathrm{geometry}}, \\ \mathsf{FreeEnergy}(T, Q) &\Longrightarrow \mathrm{RH}_{T.\mathrm{geometry}}. \end{aligned} \tag{7.11}$$

8 DIRICHLET / GRH

L_FUNCTION_POSITIVITY_PREMISE

$$\begin{gathered} N \geq 1, \quad \chi \bmod N \text{ primitive}, \\ \Lambda_\chi(s) := \Lambda(s,\chi), \quad H_\chi(s) := \Lambda_\chi(s)\Lambda_{\chi^{-1}}(s). \end{gathered} \tag{8.1}$$

$$\mathscr{D}_{\mathbf{pair}} := \texttt{completedPairCriticalSquareDomain}, \qquad \mathbb{C}_{\mathbf{slit}} := \mathbb{C} \setminus (-\infty, 0]. \tag{8.2}$$

$$\mathscr{D} := \mathscr{D}_{\mathbf{pair}} \cap \mathbb{C}_{\mathbf{slit}}, \qquad s_\pm(z) := \tfrac{1}{2} \pm \sqrt{z}. \tag{8.3}$$

$$Q_\chi(z) := |H_\chi(s_+(z))H_\chi(s_-(z))|^2 \geq 0. \tag{8.4}$$

$$\boxed{Q_\chi(z) > 0 \iff \bigwedge_{\substack{\psi \in \{\chi, \chi^{-1}\} \\ \epsilon \in \{+,-\}}} \Lambda_\psi(s_\epsilon(z)) \neq 0.} \tag{8.5}$$

$$\mathsf{LPos} :\iff \forall N \geq 1,\ \forall \chi \bmod N \text{ primitive},\ \forall z \in \mathscr{D} :\ Q_\chi(z) > 0. \tag{8.6}$$

$$\boxed{\mathsf{LFP} := \big(\mathrm{RH} \implies \mathsf{LPos}\big).} \tag{8.7}$$

$$\boxed{\mathsf{LFP} \iff (\mathrm{RH} \implies \mathrm{GRH_D}).} \tag{8.8}$$

PRIMITIVE / ALL CHARACTERS

$$\mathrm{GRH_D} := \forall N \geq 1,\ \forall \chi \bmod N \text{ primitive} :\ \mathrm{RH}_\chi. \tag{8.9}$$

$$\begin{aligned} \mathrm{GRH_{all}} :\iff & \forall N \geq 1,\ \forall \chi \bmod N,\ \forall s : \\ & [L(s,\chi) = 0,\ 0 < \operatorname{Re} s < 1] \implies \operatorname{Re} s = \tfrac{1}{2}. \end{aligned} \tag{8.10}$$

$$\boxed{\mathrm{GRH_D} \iff \mathrm{GRH_{all}}, \qquad \mathrm{GRH_{all}} \implies \mathrm{RH}.} \tag{8.11}$$

$$\begin{gathered} L(s,\chi) = L(s,\chi^*) \prod_{p \mid N} (1 - \chi^*(p)p^{-s}), \\ \operatorname{Re} s > 0 \implies \prod_{p \mid N} (1 - \chi^*(p)p^{-s}) \neq 0. \end{gathered} \tag{8.12}$$

$$\mathsf{A} \iff \mathrm{RH}, \qquad \mathsf{A} \overset{\text{OPEN}}{\implies} \mathsf{LPos}. \tag{8.13}$$

9 MERGE ALGEBRA

POSITIVITY / BINDING / CLOSURE

$$\mathfrak{A} := \{\mathsf{A}, \mathsf{R}, \mathsf{Q}_\infty, \mathsf{PM}, \mathsf{FM}_{1/2}, \mathsf{TP}, \mathsf{SL}, \mathsf{MP}, \mathsf{GSO}, \mathsf{W}, \mathsf{K}, \mathsf{C}\}. \tag{9.1}$$

$$\mathsf{Q}_\infty := \forall m > 0 : \ Q(m), \qquad \mathsf{C} := \forall \mathcal{Z}(\rho) : D(\operatorname{Re}\rho) = 0. \tag{9.2}$$

$$\boxed{\bigwedge_{X \in \mathfrak{A}} X \iff \mathrm{RH} \iff \bigvee_{X \in \mathfrak{A}} X.} \tag{9.3}$$

POSITIVITY PREMISE

$$\begin{gathered} m_i \in \mathbb{N}_{>0}, \quad \gamma_i > 0, \quad \delta_i^2 \le \tfrac{1}{4}, \\ D_i := \frac{m_i \delta_i^2 (3\gamma_i^2 + \delta_i^2)}{\gamma_i^2 (\gamma_i^2 + \delta_i^2)^2} \ge 0, \qquad c_1 := \sum_i \frac{m_i}{\gamma_i^2} < \infty, \quad C := \sum_i D_i < \infty. \end{gathered} \tag{9.4}$$

$$\mathsf{PP} :\iff \forall T \ge 1 : \ 0 \le C \le \frac{3c_1}{4T^2}, \qquad \boxed{\mathsf{PP} \iff C = 0 \iff \forall i : \delta_i = 0.} \tag{9.5}$$

ZETA BINDING / INPUT

$$\begin{aligned} \mathsf{B} :\iff \quad & \forall i : \ \mathcal{Z}(\rho_i), \quad \gamma_i = \Im \rho_i, \quad \delta_i = \Re \rho_i - \tfrac{1}{2}, \\ & \forall \rho : \ \mathcal{Z}(\rho) \Longrightarrow \exists i : \ \rho = \rho_i \ \vee \ \rho = \overline{\rho_i}. \end{aligned} \tag{9.6}$$

$$\boxed{\mathsf{B} \Longrightarrow (\mathsf{PP} \iff \mathrm{RH}).} \tag{9.7}$$

CENTRAL CURVATURE / TRACE INPUT

$$k_0 := \Re \frac{\xi''(1/2)}{2\xi(1/2)}, \qquad \mathsf{TI} :\iff k_0 = \sum_i \frac{m_i(\gamma_i^2 - \delta_i^2)}{(\gamma_i^2 + \delta_i^2)^2}. \tag{9.8}$$

$$\boxed{\mathsf{B} \wedge \mathsf{TI} \Longrightarrow (\mathrm{RH} \iff c_1 = k_0 \iff C = 0).} \tag{9.9}$$

ZHANG FIREWALL / AXIOMATIC

$$\begin{aligned} \mathsf{AX}_{\mathrm{Zhang}} \Longrightarrow \quad & \mathrm{RH}_{\mathrm{opaque}} \iff \mathsf{Hausdorff} \iff \mathsf{PascalHopf} \\ & \iff \mathsf{BKQREnergy} \iff \mathsf{TraceResolvent}. \end{aligned} \tag{9.10}$$

$$\boxed{\mathsf{LFP} \wedge \mathsf{A} \Longrightarrow \mathrm{GRH}_{\mathrm{D}}.} \tag{9.11}$$